Gastroesophageal Reflux in Infants and Children: Physiology, Natural History, and Where Reflux Becomes Reflux Disease.

Picture a three-month-old boy brought to clinic by tired parents. He brings up milk after almost every feed, five or six times a day, sometimes enough to soak his clothes. He is gaining weight along his centile, he feeds eagerly, and between feeds he is content. His parents have read that this might be reflux disease, and a relative has suggested a medicine for acid. Now picture a five-month-old girl who also spits up, but who arches her back and pulls away from the bottle, whose feeds have become a battle, and whose weight has fallen from the fiftieth to the fifteenth centile. And picture a fourteen-year-old who describes burning behind his breastbone after meals and at night.

The clinical question of this chapter is where normal gastroesophageal reflux ends and gastroesophageal reflux disease begins. Throughout, we will use the standard abbreviations: GER for gastroesophageal reflux, the physical event, and GERD for gastroesophageal reflux disease, the clinical condition. The main idea is that reflux is a normal physiological event. It is driven by a reflex that opens the lower esophageal sphincter on purpose. In infants it is frequent, mostly not acid, and self-limiting, and it follows a predictable curve that peaks at about four months and has largely disappeared by twelve to fourteen months. Disease is defined not by the reflux itself, and not by how often a baby spits, but by its consequences: troublesome symptoms or complications. The first baby in our story has physiology. The second probably has disease. The third has a condition that already looks like adult reflux disease.

A note on sources. The definitions come from the 2018 joint guideline of the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN and ESPGHAN, written by Rosen and colleagues, from the 2009 global consensus led by Sherman, from the Rome IV criteria for functional disorders in infants published by Benninga and colleagues in 2016, and from the guideline of the National Institute for Health and Care Excellence in the United Kingdom, known as NICE, first published in 2015 and updated in 2019. The physiology comes mainly from the manometry studies of Omari and colleagues in Adelaide between 1997 and 2004, the 1997 study in children by Kawahara, Dent and Davidson, and a 2006 impedance study in preterm infants by López-Alonso and colleagues. The epidemiology comes from the American practice survey by Nelson and colleagues in 1997, the Adelaide birth cohort reported by Martin and colleagues in 2002, the Italian study by Campanozzi and colleagues in 2009, and the Rome IV prevalence surveys. The treatment evidence comes from the 2009 lansoprazole trial by Orenstein and colleagues and from large cohort studies of fractures. As of 2026, no newer joint North American and European guideline on pediatric reflux has replaced the 2018 document.

Part one. The sphincter opens on purpose.

It is tempting to think of reflux as the failure of a weak valve. That picture is wrong for almost every infant and most children. The lower esophageal sphincter, the band of muscle at the junction of the esophagus and the stomach, keeps a resting tone that is usually adequate. Reflux happens mainly when the sphincter relaxes completely, briefly and without a swallow. This event is called a transient lower esophageal sphincter relaxation, usually shortened to TLESR.

The mechanism unfolds in five steps. First, a feed distends the stomach, stretching the upper part of the stomach near the cardia. Second, stretch receptors send signals up the vagus nerve to the brainstem, to a relay station called the nucleus of the solitary tract. Third, the brainstem sends a command back down the vagus that relaxes the sphincter. This relaxation is not triggered by a swallow. In the objective definition proposed by Holloway and colleagues in 1995, there is no swallow from four seconds before to two seconds after the relaxation begins, the pressure falls by at least one millimeter of mercury per second, the relaxation is complete within ten seconds, and the lowest pressure reached is two millimeters of mercury or less. Note carefully that the ten seconds is the time to complete relaxation, not the duration of the event. Fourth, with the sphincter open, the stomach and esophagus form what physiologists call a common cavity, and gastric content and gas can move up. Fifth, the esophagus clears the bolus with a swallow-induced peristaltic wave, saliva neutralizes what is left, and in infants milk itself buffers much of the acid.

Why does such a reflex exist at all? Because the same transient relaxation is how we vent swallowed air. Belching is its physiological purpose. In an infant who swallows air with every feed, frequent relaxations are part of normal life.

Part two. Most reflux rides on a transient relaxation.

The evidence for this mechanism in infants is strong and remarkably consistent. In Omari's 1997 study of thirteen healthy preterm infants, ninety-four percent of reflux episodes occurred during transient relaxations. In a second study of twenty-four healthy preterm infants published in 1998, eighty-two percent of one hundred and ninety-three reflux episodes were linked to a transient relaxation. Those relaxations lasted on average about fifteen seconds, compared with under six seconds for an ordinary swallow-induced relaxation, and the lowest pressure reached was below one millimeter of mercury, compared with more than two. In a 2004 study of body position, eighty-three percent of reflux in preterm infants was triggered this way.

The most instructive study is Omari's from 2002, which compared infants with reflux disease with controls. Transient relaxations triggered a median of about ninety-one and a half percent of reflux episodes. But infants with reflux disease did not have more transient relaxations than controls. What differed was that a larger share of their relaxations carried acid reflux, sixteen and a half percent compared with about six percent. In other words, the disease was not a different mechanism. It was the same physiological event with a different content and a different consequence.

Older children show a similar but more mixed picture. In the 1997 study by Kawahara and colleagues, transient relaxations explained fifty-eight percent of reflux episodes in children with pathological reflux and sixty-nine percent in children without it. Swallow-associated relaxations

explained about a fifth more. Persistently absent sphincter tone, the weak valve of the old picture, explained only eight percent of episodes, and only in the children with pathological reflux.

There is a therapeutic lesson here. Acid suppression does not touch the mechanism. It changes what refluxes, not whether reflux happens.

Part three. Why infants are built to reflux.

Four features of early life make reflux more frequent and more visible. The first is anatomy. The segment of esophagus below the diaphragm is short, the angle at which the esophagus enters the stomach, called the angle of His, is not yet acute, and the esophagus itself holds only a small volume, so even a small reflux reaches the mouth. The second is diet. Infants live on liquid milk, fed frequently and in large volumes relative to their size, and overfeeding makes regurgitation worse. The third is position. Infants spend much of their time lying down. In Omari's 2004 positioning study, lying on the right side increased reflux but sped gastric emptying, while lying on the left side reduced reflux. These findings explain physiology, but they do not change safe sleep advice: infants should sleep on their backs because of the risk of sudden infant death.

The fourth feature, buffering, deserves its own discussion, because it changes how we measure reflux.

Part four. Most infant reflux is not acid.

Milk is a buffer. In a 2001 study by Mitchell and colleagues, after a feed the infant stomach stayed above a pH of four for an average of about one hundred and thirty minutes. The reflux index, the percentage of time the esophageal pH probe reads below four, was on average one percent. But when the authors counted only the periods when the stomach was actually acid, the index rose to about two and a half percent. Their conclusion was that a low reflux index may reflect prolonged buffering of gastric acid rather than the absence of reflux.

Impedance monitoring, which detects the movement of any liquid or gas bolus regardless of its acidity, confirms this. In López-Alonso's study of twenty-one healthy preterm neonates, there was a median of seventy-one reflux events in twenty-four hours. About a quarter of them were acid, about three quarters were weakly acidic, meaning a pH between four and seven, and none were weakly alkaline. Ninety percent of the episodes reached the upper esophagus, and the stomach was above a pH of four for about seventy percent of the recording.

For the clinician, this means that a pH probe alone sees only a fraction of infant reflux. A normal pH study does not mean that reflux is absent, and an abnormal one does not by itself mean that the reflux is causing the symptoms.

Part five. The curve that reassures parents.

The natural history of infant regurgitation is one of the most useful things to know in pediatric gastroenterology, because showing it to parents is often the most effective treatment. In Nelson's 1997 survey of nine hundred and forty-eight parents in nineteen American practices, about half of

infants aged zero to three months regurgitated at least once a day. The peak came at four months, when sixty-seven percent did so. At six months the figure was still sixty-one percent, but it then fell sharply to twenty-one percent at seven months and to five percent at ten to twelve months. Interestingly, parents were most likely to call the regurgitation a problem at six months, twenty-three percent of them, just as it was about to resolve. And in 1997, only two in a thousand of these infants were given medication.

The Adelaide birth cohort reported by Martin and colleagues tells the same story. Spilling of most feeds peaked at forty-one percent at three to four months and fell below five percent by thirteen to fourteen months. In the Italian study by Campanozzi and colleagues, fifty-nine pediatricians examined two thousand six hundred and forty-two infants, of whom twelve percent met the Rome II criteria for infant regurgitation. Of the two hundred and ten infants followed, about twenty-seven percent had resolved by six months, about eighty-eight percent by twelve months, about ninety-nine percent by eighteen months, and all of them by twenty-four months. Only one of the two hundred and ten, half a percent, was eventually diagnosed with reflux disease. Breastfed infants resolved a little earlier than formula-fed infants, at about eight months against nine and a half. Both the Rome IV committee and NICE give the same message: about nine in ten infants have stopped regurgitating by their first birthday.

Part six. Not entirely benign.

There is one caveat to the reassurance. The Adelaide cohort followed six hundred and ninety-three of its eight hundred and thirty-six children, eighty-three percent, to about nine years of age. Children who had spilled frequently for ninety days or more in infancy were more likely to have reflux symptoms at nine. The relative risk was about two point three for any reflux symptom, about four and a half for heartburn, about two point seven for vomiting and about four point seven for acid regurgitation. The confidence intervals were wide, but all excluded no effect.

How should we read that? Prolonged infant spilling identifies a child at somewhat higher later risk. It is a marker. It does not show that treating infant regurgitation would prevent later disease, and no study has tested that.

Part seven. The number depends on the definition.

Prevalence figures for infant regurgitation vary widely, and the main reason is the definition. Nelson counted daily regurgitation. The Rome IV criteria, which we will describe in detail later, require at least two regurgitations a day for at least three weeks. Under Rome IV, about fourteen percent of nearly seventeen hundred European infants qualified, about twenty-four percent of American infants in an online survey of mothers, about thirteen percent in Turkey and about ten and a half percent in Malaysia. A Chinese survey found about thirty-four percent in infants up to six months but only about three and a half percent between seven and twelve months, which is the natural history curve again. In the European cohort, formula feeding was associated with Rome IV regurgitation. The practical point is to quote the definition along with the number, because sixty-seven percent and fourteen percent can both be correct.

Part eight. After infancy, symptoms turn adult-like.

Beyond infancy, reflux symptoms look more like those of adults. In Nelson's 2000 survey, parents of children aged three to nine reported heartburn in about two percent, epigastric pain in about seven percent and regurgitation in about two percent. Parents of children aged ten to seventeen reported heartburn in three and a half percent, epigastric pain in three percent and regurgitation in about one and a half percent. When the adolescents answered for themselves, the figures were higher: heartburn in about five percent, epigastric pain in five percent and regurgitation in about eight percent, roughly six times what their parents reported for regurgitation. Heartburn in adolescents was associated with cigarette smoking, with an odds ratio of six and a half. Reviews generally quote reflux disease symptoms in roughly two to eight percent of children and adolescents. The lesson is to ask the adolescent directly.

Part nine. Where reflux is disease from the start.

In some children the reassuring curve does not apply. After repair of esophageal atresia, reflux is the commonest gastrointestinal complication, reported in about twenty-two to forty-five percent, and the 2016 joint guideline on esophageal atresia recommends acid suppression from the neonatal period through about the first year. In children with neurological impairment, vomiting, regurgitation or rumination affects about twenty to thirty percent, and after fundoplication, the operation that wraps the top of the stomach around the lower esophagus, reflux recurs more often than in other children, about fourteen percent against nine percent, with redo surgery needed in about thirteen percent against nine percent. In children with congenital anomalies such as congenital diaphragmatic hernia, abdominal wall defects and malrotation, narrative reviews report reflux disease in half or more, although these figures come from small series. And in preterm infants the mechanism is the same transient relaxation, the reflux is mostly non-acid, and it is rarely the cause of apnea or slow heart rate on its own. In all these groups reflux persists, complications such as esophagitis, aspiration and stricture are more common, and the threshold for investigation is lower.

Part ten. Four documents, one boundary.

Now to the boundary itself. The 2018 NASPGHAN and ESPGHAN guideline defines GER as the passage of gastric contents into the esophagus with or without regurgitation and vomiting. It defines GERD as reflux that leads to troublesome symptoms affecting daily functioning, complications, or both, and it calls GERD refractory when it does not respond to optimal treatment after eight weeks. The 2009 global consensus defined GERD as the symptoms and complications that develop secondary to persistent reflux, and it made the important point that regurgitation is neither necessary nor sufficient for the diagnosis. NICE defines reflux disease as reflux that causes symptoms severe enough to merit medical treatment, or complications such as esophagitis or aspiration. Each wording is slightly different, but every one places disease in consequences, never in the number of spits.

Rome IV takes a different approach for infants. Infant regurgitation, in Rome IV, applies to an otherwise healthy infant between three weeks and twelve months of age who regurgitates at least

twice a day for at least three weeks, without retching, vomiting blood, aspiration, apnea, failure to thrive, feeding or swallowing difficulty, or abnormal posturing. Rome IV deliberately left out the word troublesome. So the same frequent spitter without red flags is a functional infant regurgitation in Rome IV regardless of how the parents feel, whereas NASPGHAN and ESPGHAN would call the same infant GERD if the symptoms were troublesome.

The red flags matter more than the label. Bilious vomiting, onset after six months or persistence beyond twelve to eighteen months, vomiting of blood, fever, lethargy, a bulging fontanelle, an abnormally large or small head, abdominal distension, chronic diarrhea and weight loss all send you to a differential diagnosis, which includes pyloric stenosis, malrotation, infection, raised intracranial pressure, metabolic disease and food allergy, not to an acid suppressant.

Part eleven. Crying is not reflux.

The commonest error in this field is to treat crying as if it were reflux. The 2018 guideline states that if excessive irritability and pain is the single manifestation, it is unlikely to be related to reflux disease, and that a short trial of a proton pump inhibitor, the class of strong acid-suppressing drugs that includes omeprazole and lansoprazole, is not recommended as a diagnostic test in infants.

The evidence behind that is clear. In the trial by Orenstein and colleagues, one hundred and sixty-two infants with symptoms attributed to reflux disease were randomized to lansoprazole or placebo. Fifty-four percent responded in each arm, an identical result. Serious adverse events were more common with lansoprazole, ten against two, a statistically significant difference driven mainly by lower respiratory tract infections. In an earlier trial by Moore and colleagues, omeprazole reduced the reflux index far more than placebo, by about nine percentage points against two, yet irritability did not change, and crying fell over time in both groups. A systematic review in 2015 concluded that the randomized trials do not support using these drugs to reduce infant crying and irritability. The drug changed the acid; it did not change the baby.

Part twelve. Measuring the boundary.

Can a test find the boundary? Only partly. The 2018 guideline gives pediatric pH thresholds: a reflux index above ten percent is abnormal under one year of age, and above five percent over one year. But when compared with history and examination, the pH index had a sensitivity of about fifty percent and a specificity of about eighty-two percent, which is modest. In adults, the Lyon Consensus 2.0, published in 2024, defines conclusive reflux disease as an acid exposure time above six percent and excludes it below four percent, uses more than eighty reflux episodes a day as supportive evidence and fewer than forty as evidence against, and uses a mean nocturnal baseline impedance below fifteen hundred ohms as supportive evidence and above twenty-five hundred ohms as evidence against. It also treats erosive esophagitis of Los Angeles grade B or worse as conclusive. The Lyon document does not mention children, and infants have more reflux episodes than older children, so its episode counts cannot be borrowed. The finding of significant erosive esophagitis is the one element that clearly transfers. Most importantly, normal values for pediatric impedance have never been established; the European pediatric protocol from 2012 rests on expert consensus, and the large 2011 cohort by Pilic and colleagues described symptomatic

children, not normal ones. In children, impedance is read mainly through the association between symptoms and reflux events rather than against a number.

Part thirteen. The physiology is being treated.

Despite all of this, acid suppression in infants has grown. In a French national study of more than eight million children born between 2010 and 2021, about seven hundred thousand, roughly eight and a half percent, received a proton pump inhibitor before their first birthday, and the incidence of such prescriptions without any hospital diagnosis of reflux disease rose by about sixty-four percent over the period. The harms are not trivial. In an American cohort of about eight hundred and fifty thousand children, eleven percent received acid suppression in their first year, and the hazard of later fracture was about twenty-one percent higher with a proton pump inhibitor alone and thirty percent higher with a proton pump inhibitor plus an H2 receptor antagonist, the older class that includes famotidine. Fracture risk rose with earlier and longer exposure, while an H2 receptor antagonist alone carried no increase. A Swedish matched cohort found an overall hazard ratio of about one point one, rising to about one point three for a year or more of use. These are observational studies, and confounding by indication is possible. But in infants with uncomplicated regurgitation or crying, the benefit side of the ledger is empty.

Part fourteen. What the guidelines say.

On where reflux disease begins, the guidelines differ in wording: troublesome symptoms or complications in NASPGHAN and ESPGHAN, symptoms severe enough to merit treatment in NICE, and no troublesome criterion at all in Rome IV. On isolated regurgitation, they are aligned: reassure, and do not give acid suppression; NICE states explicitly that these drugs should not be offered for overt regurgitation as an isolated symptom. On crying or irritability alone, they are aligned as well: NASPGHAN and ESPGHAN call it unlikely to be reflux disease, and NICE advises against investigating or treating distressed behavior without overt regurgitation. On a diagnostic trial of acid suppression, they are aligned for infants, while NASPGHAN and ESPGHAN allow a four to eight week trial in older children with typical symptoms such as heartburn.

Where they part is the step before any acid suppressant. NASPGHAN and ESPGHAN suggest feeding changes and then a two to four week trial of an extensively hydrolyzed or amino acid formula, because cow's milk protein allergy can mimic reflux disease. NICE recommends smaller, more frequent feeds, then thickened formula, and an alginate, a raft-forming agent, for one to two weeks. A 2026 Italian consensus allows a brief trial of any of these. Finally, on adult pH-impedance criteria in children, there is a gap: none of the pediatric documents adopts them, and the adult consensus does not address children.

Part fifteen. Traps and pearls.

Four traps recur. The first is treating crying as reflux: irritability is the only symptom, a proton pump inhibitor is started, there is no benefit and there are more infections. The second is reassurance from a normal pH study recorded after milk feeds, when the acid was buffered for two

hours and the non-acid reflux was invisible to the probe. The third is calling everything just reflux and never asking about red flags, so that bilious vomiting, pyloric stenosis or malrotation is missed, or vomiting that began after six months is not questioned. The fourth is assuming that reflux will settle in a child after esophageal atresia repair or with neurological impairment, where the natural history does not apply. And one more: labelling a child's impedance study normal or abnormal using adult Lyon numbers, when pediatric normal values do not exist.

The numbers worth carrying are these. Transient relaxations carry about eighty-two to ninety-four percent of reflux in infants, and infants with reflux disease do not have more of them, only more acid in each. The transient relaxation is a vagal reflex triggered by gastric distension, and it also vents swallowed air. About three quarters of preterm reflux is weakly acidic, and milk keeps gastric pH above four for about two hours. Daily regurgitation affects about half of infants at zero to three months, sixty-seven percent at four months and five percent at ten to twelve months. About nine in ten have resolved by twelve months, and in the Italian study only one of two hundred and ten regurgitating infants had reflux disease. Frequent spilling for ninety days or more predicted heartburn at nine years with a relative risk of about four and a half. Reflux disease means troublesome symptoms or complications, and regurgitation alone is neither necessary nor sufficient. Lansoprazole and placebo produced fifty-four percent responders each, with ten serious adverse events against two. The pediatric reflux index is abnormal above ten percent under one year and above five percent over one year, and adult Lyon thresholds are not validated in children. And children with repaired esophageal atresia, neurological impairment or major congenital anomalies have reflux disease until proven otherwise.

What to do differently in clinic.

Let us return to the three children. The three-month-old who spits five or six times a day but is thriving and content has physiological reflux. The most useful thing to do is to explain the mechanism and show the parents the curve: two thirds of babies do this at four months, and nine in ten have stopped by their first birthday. Check that he is not being overfed. No tests and no drugs are needed. If he also cried after feeds but kept growing, the crying alone would not make this reflux disease; a feeding review, and perhaps a brief trial of thickened feeds, an alginate or a milk-protein-free formula, would come before any thought of acid suppression.

The five-month-old who spits, arches, refuses feeds and is falling across centiles has troublesome symptoms with a complication, failure to thrive. She probably has reflux disease, but first she needs a careful search for red flags and for other causes, including cow's milk protein allergy. Treatment is then targeted and time-limited, and reassessed. A seven-month-old with new projectile or bilious vomiting does not have reflux until proven otherwise. An eighteen-month-old who still regurgitates every day is outside the natural history and deserves evaluation for anatomical problems, eosinophilic esophagitis and rumination. An infant after esophageal atresia repair should receive acid suppression through the first year and be followed.

The fourteen-year-old with recurrent heartburn has adult-type reflux disease. A four to eight week trial of a proton pump inhibitor is reasonable, along with advice to stop smoking, and if symptoms

persist or alarm features appear, endoscopy is the next step.

So, in clinic tomorrow: think of reflux as a reflex, not a broken valve. Show parents the curve. Define disease by consequences, not by the number of spits. Never treat crying alone with acid suppression. Ask the adolescent, not only the parent. Do not borrow adult thresholds to read a child's pH-impedance study. And keep the reassuring curve for healthy infants; in children with esophageal atresia, neurological impairment or major anomalies, expect disease and look for it.